

EDWARD ILOEMEZUE

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Professional Summary

Mechanical Engineering graduate with hands-on experience in CAD design, prototyping, and mechanical system development. Skilled in SolidWorks modeling, CFD analysis, and translating design concepts into fabricated components through CNC machining and rapid prototyping. Experienced in technical documentation, testing, and cross-functional collaboration. Detail-oriented problem solver with a strong interest in developing innovative mechanical systems in fast-paced, multidisciplinary environments.

Education

Master of Engineering in Mechanical Engineering
University of Ontario Institute of Technology | Oshawa, Ontario
January 2025 – June 2026

Bachelor of Engineering in Mechanical Engineering
Covenant University | Ogun, Nigeria
September 2018 – August 2023

Core Qualifications

- CAD Proficiency
- MATLAB/Simulink
- C++ Programming
- Microsoft Office Proficiency

Industry Experience

- Pulita Energy | Lagos, Nigeria
Technical Engineer
Sept 2024 – Dec 2024

 - Conducted generator inspections and preventive maintenance to improve asset reliability and reduce operational downtime across multiple sites.
 - Enhanced project decision-making by delivering technical presentations to stakeholders with clear data-driven recommendations.
 - Collaborated with operations teams to align technical designs with project objectives.
- Kinetics Africa | Lagos, Nigeria
Technical Intern
Jan 2024 – Aug 2024

 - Increased alignment between design and operations by collaborating with cross-functional teams on generator projects.
 - Improved engineering documentation accuracy by contributing to reports and records used for project planning
 - Provided data analysis support that enabled engineers to optimize operations.
- Mikano Steel Factory | Lagos, Nigeria
Design Engineer
May 2022 – Sept 2022

 - Accelerated production timelines by creating precise CAD designs and Bills of Materials (BOM) for generator canopies and industrial equipment.
 - Reduced fabrication errors through accurate engineering drawings that guided the assembly team.

Metropolitan Motors | Lagos, Nigeria
Intern
Sept 2021

- Improved manufacturability by resolving design challenges in steel equipment using SolidWorks and AutoCAD.
- Increased workshop efficiency by diagnosing automotive faults with computerized tools, reducing repair turnaround time.
- Supported senior technicians in performing preventive maintenance and engine overhauls to ensure vehicle reliability.
- Contributed to workplace safety and efficiency by maintaining an organized repair environment.

Additional Experience

Real Canadian Superstore | Oshawa, Canada
Clerk Part Time
Aug 2025 - Present

- Supported operational efficiency through inventory management, shelf stocking, and standardized food preparation processes.
- Maintained accurate stock levels and organized workflows to ensure smooth daily operations.

SportChek | Oshawa, Canada
Store Advisor – SC Footwear Part Time
Oct 2025 - Present

- Advised customers on footwear selection by assessing needs, fit, and performance requirements, supporting confident purchasing decisions.
- Maintained up-to-date product knowledge on footwear technologies and new releases, delivering clear and effective customer communication.

Project Experience

PID-Based Control Optimization of Robotic Manipulator

Advanced Optimization – University of Ontario Institute of Technology | May 2025 – June 2025

Software: MATLAB, Simulink

PID-based control system for a robotic manipulator was optimized using numerical techniques in MATLAB/Simulink. Controller gains were tuned to minimize error, improve response, and enhance stability, with simulations validating the optimized controller's robustness.

Generator System Performance & Reliability Optimization

Pulita Energy | Nigeria | September 2024 – December 2024

Analyzed generator data, load profiles, and maintenance records to identify inefficiencies and risks. Supported commissioning, inspections, and preventive maintenance planning, contributing to improved asset availability and reduced downtime.

Engineering Design & Manufacturing Optimization of Steel Equipment

Mikano Steel Factory | Lagos, Nigeria | May 2022 – September 2022

Software: SolidWorks, AutoCAD

Designed generator canopies and industrial steel equipment using SolidWorks and AutoCAD. Collaborated with production teams to optimize designs for manufacturability and accuracy.

Certifications

- Certified Solidworks Associate- Mechanical Design (CSWA- Mechanical Design)
- Certified Solidworks Professional- Mechanical Design (CSWP- Mechanical Design)
- Certified Solidworks Advanced Professional Sheet Metal (CSWPA-SM)
- Certified Solidworks Advanced Professional Weldments (CSWPA-WD)
- Certified Solidworks Advanced Professional Drawing Tools (CSWPA-DT)